

**TOPOGRAPHIC CHARACTERIZATION AND EVOLUTIONARY MODELING OF LARGE VOLCANO-TECTONIC STRUCTURES ON VENUS.** Alexa K. Schultz<sup>1,2</sup> and Patrick J. McGovern<sup>1</sup>, <sup>1</sup>Lunar and Planetary Institute (USRA), Houston, TX 77058, <sup>2</sup>Yale University Department of Earth and Planetary Sciences, New Haven, CT 06510.

**Introduction:** The surface of Venus is dominated by volcanic features; there are over 80,000 volcanoes on Venus as detected by radar imaging from the Magellan mission in the early 1990s [1]. The largest such structures (150 or so), termed “large volcanoes” or large shields [2], are analogs to intraplate volcanoes such as Kilauea on Earth. However, another comparably large class of volcano-tectonic landforms, the corona, is defined by the International Astronomical Union simply as an “ovoid-shaped feature” [3]. This term has taken on new meaning since its original conception, with the now-common assumption within the Venus community is that most corona morphology is generated via the lithospheric response to upwelling mantle material [e.g., 4].

Here, we take a quantitative approach to characterizing the topography of large volcano-tectonic structures on Venus, applying harmonic analysis in axisymmetry – the Hankel transform – to objectively evaluate affinities and differences between Venus volcanic structures of various classes and to evaluate the utility of existing classification schemes. We also calculate quantitative models of evolving volcanic edifice topography, in order to evaluate the conditions that contribute to producing the wide range of shapes of volcano-tectonic structures on Venus.

**Methods:** *Data analysis.* The base Venus topography dataset is the Global Topographic Data Record GTDR-SINUS.3;2 [5]. We determine the topographic signatures of specific volcano-tectonic constructs on Venus, spanning the classifications “Mons” and “Corona” [2, 4], using the pyGMT mapping package [6]. We collect topographic cross-sections along 24 distinct azimuths (15° intervals) at 5 km spacing out to 450 km. These are averaged to create a mean topographic profile.

We use the Hankel transform, the axisymmetric analog of Fourier series, to analyze the topographic signatures of large volcanic constructs on Venus. Following [7], we use a trapezoidal integration scheme to integrate the forward and inverse Hankel transforms, for example, representing a function via a sum of power spectrum coefficients  $h_n$  multiplied by scaled zeroth-order Bessel functions:

$$h(r) = \sum_{n=1}^{nk} h_n J_0(k_n r) \quad (1)$$

The wavenumber  $k_n$  values are evenly distributed across the desired spectral band, and the  $h_n$  coefficients are determined by trapezoidal integration.

We obtain the following quantities from the topographic power spectra of Venus volcanic

structures: the wavelength ( $\lambda_L$ ) and power ( $P_L$ ), the wavelength ( $\lambda_{\max}$ ) and power ( $P_{\max}$ ) at the maximum topographic power peak, the wavelength ( $\lambda_{\min}$ ) and power ( $P_{\min}$ ) at the minimum topographic power peak, and the wavelength ( $\lambda_A$ ) and power ( $A_{\max}$ ) at the absolute maximum topographic power peak. We then define four non-dimensional parameters [e.g., 8] defining several aspects of the volcanic topography.

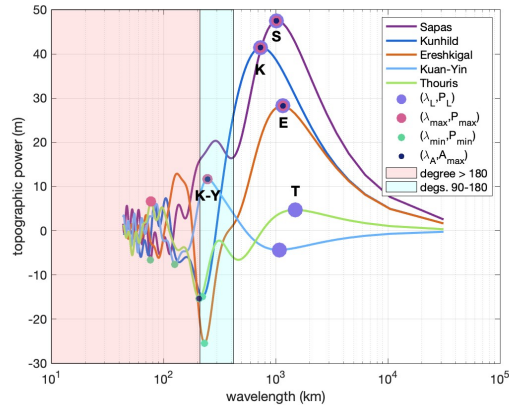
$$\Pi_1 = A_{\max}/P_L \quad (2)$$

$$\Pi_2 = |P_{\min}/P_L| \quad (3)$$

$$\Pi_3 = \lambda_A/\lambda_L \quad (4)$$

$$\Pi_4 = \lambda_{\min}/\lambda_L \quad (5)$$

$\Pi_1$  describes edifice nature,  $\Pi_2$  describes annular nature,  $\Pi_3$  describes the dominant spectral peak nature, and  $\Pi_4$  describes the importance of a short wavelength spectral minimum.



**Figure 1.** Topographic power spectra [7] of 5 structures on Venus, showing the longest wavelength peak (purple dot), maximum topographic power peak (pink dot), minimum topographic power peak (teal dot), and absolute maximum topographic power peak (navy dot). Sapas Mons is a conical volcano, Kunhild, Ershkigal, and Kuan-Yin Coronae are annular structures, and Thouris Corona is a hole-like structure. Blue bar shows the expanded wavelength band of gravity field resolution expected from VERITAS [9].

**Modeling.** We model self-consistent growth of volcanic edifices on Venus by calculating the interaction of the lithospheric stress state generated by lithospheric flexure with favored magma ascent pathways [10-12]. We use an iterative method [13] that 1) uses stress-based magma ascent criteria to determine locations of eruption, 2) distribute material via lateral flow, 3) applies the resulting edifice topography (pre-existing plus new increment) as a lithospheric load, generating new topography and stress state, and 4) iterates until all load increments are emplaced. Model

input parameters include lithospheric thickness  $T_e$ , magma source radius ( $r_0$ ), central height ( $h_0$ ), and shape (cone, truncated cone, or annulus), and the number of load increments  $n_{inc}$  [11-13].

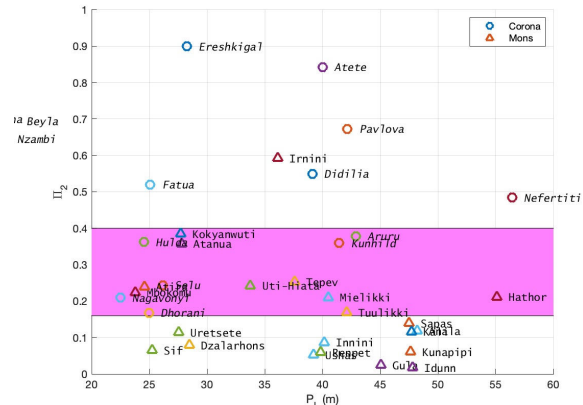
**Results and Discussion:** We analyzed the topographic signatures of 49 volcano-tectonic features on Venus; results for three characteristic shapes are shown in Figure 1. Sapas Mons is a conical edifice with a very strong topographic power peak of about 48 m at 1000 km wavelength. The power spectrum declines rapidly as wavelength decreases. For Kuan-Yin Corona, an annular structure, the longest wavelength peak has negative topographic power, and the highest magnitude peak occurs at a shorter wavelength and lower topographic power than that of Sapas Mons. Kunhild and Ereshkigal Coronae show long-wavelength power spectrum peaks similar to that of Sapas Mons (Fig. 1), but with the addition of shorter-wavelength minima that generate the central lows of their annular topographic profiles. These findings are consistent with a view of Kunhild and Ereshkigal Coronae as features with affinities to both shield volcanoes (in terms of origin) and coronae (in terms of topographic shape) [14], and illuminate a “continuum” or “spectrum” of volcanic edifice shapes spanning these structures. The wavelength band containing many of these distinctive spectral features is not resolved in current gravity field solutions but may be by VERITAS radio science [9].

Thouris Corona represents a class of feature with deep central depressions and minimal surrounding topography, and with maximum absolute magnitude topographic power negative in sign (opposite to Sapas and Kuan-Yin) and at a short wavelength (similar to Kuan-Yin but opposite to Sapas).

We use the topographic power spectra information to characterize the topography of large volcano-tectonic constructs on Venus. For example, examination of the  $\Pi_2$  parameter (Figure 2) demonstrates three characteristic groupings: features with classifications mons and corona both appear in a band  $0.16 < \Pi_2 < 0.4$ . For lower values, features designated montes dominate, whereas for higher values, features designated coronae dominate.

An x-y plot of the components of the  $\Pi_1$  parameter reveals a “main sequence” of 39 features along the 1:1 line, on which 33 features classified mons and corona are intermixed with no discernable pattern. Our models of volcanic edifice construction are able to replicate the morphology of some coronae on the 1:1 line, including Kuan-Yin and Fatua. Thus, we suggest that

such structures could be volcanic edifices dominantly constructed by surface flows, even those that have been given the classification “corona” – these could be considered “strangely shaped volcanic edifices” [10]. Six features with a deep central depression, classified as coronae, plot below the 1:1 line. We suggest that they likely have a different formation mechanism than coronae that lie on the 1:1 line.



**Figure 2.** Non-dimensional parameter  $\Pi_2$  versus topographic power at longest wavelength peak ( $P_L$ ) for volcanic features classified as corona (circles) and mons (triangles). Pink bar denotes range of  $\Pi_2$  values transitional between mons-dominated and corona-dominated values.

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